1. ER diagram

ER diagram A driving school

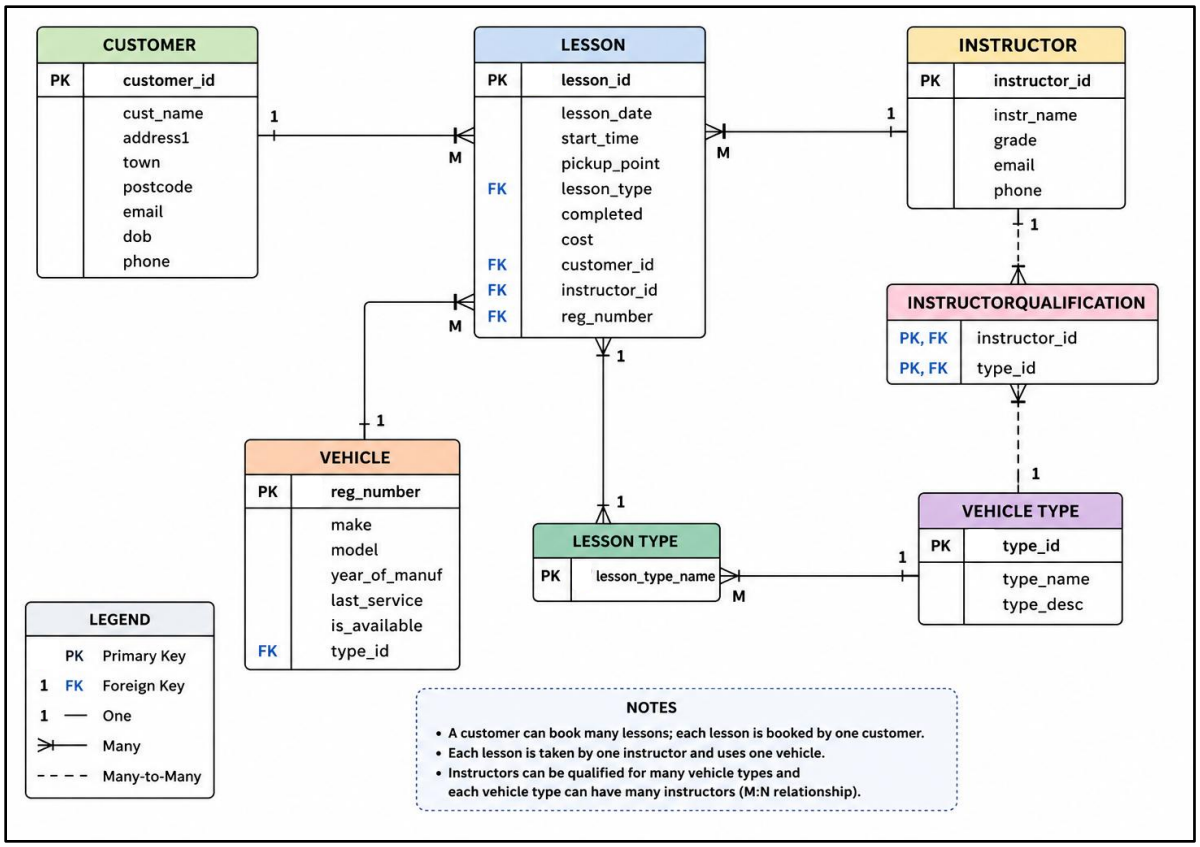

Figure 1: Entity Relationship Diagram of A driving school

Figure 1 displays the ER diagram for a driving school management system. The main entities are Customer, Instructor, Vehicle, VehicleType, Lesson, LessonType, and InstructorQualification. Vehicle refers to training cars; Lesson records driving experiences; Customer saves learner information; and teacher saves teacher information. VehicleType and LessonType describe vehicle and lesson categories. InstructorQualification is a junction object that links vehicle types and instructors.

While each object has a primary key, InstructorQualification uses a composite key (instructor_id, type_id). Foreign keys are used to preserve relationships like Vehicle–VehicleType and Lesson–Customer, Instructor, Vehicle, and LessonType.

.

2: Relational Database Schema (3NF Design)

VehicleType (type_id PK, type_name, type_desc)

Vehicle (reg_number PK, make, model, year_of_manuf, last_service, is_available DEFAULT 1,

type_id FK → VehicleType(type_id) ON DELETE RESTRICT ON UPDATE CASCADE)

Instructor (instructor_id PK, instr_name, grade, email, phone)

InstructorQualification (instructor_id PK FK → Instructor(instructor_id),

type_id PK FK → VehicleType(type_id))

Customer (customer_id PK, cust_name, address1, town, postcode, email, dob, phone)
LessonType (lesson_type_name PK)

Lesson(lesson_id PK, lesson_date, start_time, pickup_point,

lesson_type FK → LessonType(lesson_type_name) DEFAULT 'Standard',

completed DEFAULT 0, cost,

customer_id FK → Customer(customer_id) ON DELETE RESTRICT,

instructor_id FK → Instructor(instructor_id) ON DELETE SET NULL,

reg_number FK → Vehicle(reg_number) ON DELETE SET NULL)

Normalization (3NF)

The schema satisfies Third Normal Form (3NF):

- All attributes are atomic (1NF)
- No partial dependency exists (2NF)
- No transitive dependency exists (3NF)

Constraints and Design Features

- Primary keys guarantee that each record is uniquely identified.
- Referential integrity is preserved via foreign keys.
- Data consistency is enhanced by default values (finished = 0, is_available = 1).
- ON DELETE rules prevent improper deletions (RESTRICT) and also help to safeguard important data by setting it to NULL (SET NULL).
- A junction table resolves many-to-many relationships (InstructorQualification).

Optimisation and Scalability

- Use of lookup tables (LessonType, VehicleType) reduces redundancy
- Junction table design allows flexible expansion
- Efficient key structure improves query performance
- Schema is extensible for future features such as pricing or scheduling

3: Limitations and Assumptions

The main presumptions established throughout the design process and the shortcomings of the existing database design are described in this section.

Assumptions

- Only one client makes reservations for each lesson.
- One instructor and one car are used for each lesson.
- Since they may be assigned later, the instructor and vehicle may be NULL at the time of booking.
- The scenario states that each lesson has a two-hour time limit.
- The lesson start time is recorded as an integer (hour format) for convenience.
- Every vehicle is a member of a single vehicle type.
- It is possible for a teacher to be qualified to train different kinds of vehicles.
- The availability of a vehicle is indicated by a binary number (1 or 0).
- Each lesson must fit into one of the authorized lesson categories, such as Advanced or Standard.
- All entered data, including phone numbers and emails, is thought to be authentic and properly formatted.

Limitations

- Minutes and seconds are not supported because the system only stores time in whole hours.
- Only instructor multiple booking is prevented; there is no conflict checking for car reservations.
- A feature for rescheduling or canceling lessons is absent from the design.
- Redundancy could result from the incomplete normalization of customer addresses.
- There is no system for user login or authentication.
- The system doesn't keep track of past instructor changes or vehicle maintenance.
- The cost of the lesson is input manually rather than computed automatically.
- Pickup locations are not stored with structured location data; instead, they are stored as free text.
- Large-scale systems and high concurrency are not supported by the database.
- Some data validation (e.g., email format, date checks) is handled outside the database.